DATA SCIENCE 1 · 2026–2027

Combine, summarize, reshape

Syllabus for lab 4 – *Data Processing II*

Lecturer	dr. Maarten Slembrouck
Session	lab 4 – *Data Processing II*
Materials	`https://github.com/PACE-Ghent/Data-Science-1-2026`

1 Overview

In labs 3, 4 and 5 you are the new data analyst of a (fictional) youth football and athletics club. The head coach gives you three export files from the club's systems. They are messy on purpose, the way real exports are: inconsistent labels, sprint times in milliseconds, missing values, an athlete who appears twice and a column with a comma as the decimal separator.

Each session builds on the previous one:

Lab 3 **From file to tidy table.** Load, explore, filter, add columns and clean until you can trust the data.

Lab 4 **Combine, summarize, reshape.** Summarize groups, combine tables, switch between a wide and a long shape, and work with a heart-rate series per second.

Lab 5 **Tell the story with plotly.** Choose the right chart for a question, make honest charts with `plotly.express` and combine them into a small interactive dashboard.

2 Learning outcomes

After lab 4 you can:

LO1 summarize data per group with `groupby` and `agg`, and combine tables with `merge` without double counting;

LO2 reshape data between a long and a wide shape with `melt` and `pivot_table`;

LO3 compute basic features of a time series: heart-rate zones, rolling averages and resampling.

3 Prerequisites

Lab 3 (*Data Processing I*): loading a CSV file with `pandas`, selecting and filtering rows and columns, computing new columns and cleaning the data. Lab 4 starts with the cleaning pipeline of lab 3, in compact form.

4 Software and materials

Every lab is a **marimo notebook** in the `labs/` folder of the course repository. You can open it in two ways:

- **Online:** go to `https://marimo.app`, choose *New* → *Open from URL...* and paste the raw URL of the notebook on GitHub. The data is then read from GitHub, so you don't have to install anything.

- **Locally:** install `uv` and Python 3.12 following the *Installing Python* manual (or the slides). Then go to the course folder in a terminal and open a lab with:
  `uv run marimo edit labs/lab3_data_processing_I.py`
  The first time, uv installs the packages it needs; the data is read from the `data/` folder.

Three things to know about marimo:

- Notebooks are **reactive**: when you change a cell, every cell that depends on it runs again, in the order of their dependencies.
- **Two cells cannot define the same variable.** Give every step its own clear name, such as `tests_raw` and `tests_clean`, or keep a sequence of steps together in one cell.
- Every exercise contains a `...` placeholder. A hidden cell below it checks your answer: it turns **green** when the answer is right, and otherwise gives a general hint about what is still wrong, without giving the answer away.

5 The dataset

File	Size	Content and known problems
`athletes.csv`	43 rows, 7 columns	ID, name, birthdate, gender, position, height, weight. 13 spellings for 4 positions, weight as text with a decimal comma (`"57,5"`), one athlete entered twice with two IDs, a few missing values.
`tests.csv`	114 rows, 8 columns	10 m and 30 m sprint, countermovement jump height and Yo-Yo level, at 3 test moments. Some sprint times are stored in milliseconds; a few missing values.
`hr_session.csv`	12,000 rows, 5 columns	Heart rate and speed per second for 5 athletes during one 40-minute training session.

6 Schedule

Goal: aggregate data with `groupby`, combine tables with `merge`, reshape with `melt` and `pivot`, and compute basic features of a time series.

Time	Block	What you practise
13:00–13:15	Recap: repair a broken cell	Fix a small cleaning pipeline with two bugs.
13:15–14:10	`groupby` + `agg`	Statistics per group, age groups with `pd.cut`, several named aggregations at once.
14:10–14:55	`merge`	Inner and left joins; why a duplicated athlete is counted twice, and why you remove duplicates before merging.
14:55–15:10	Break	
15:10–15:50	`melt` and `pivot`	Long and wide shape; progress between two test moments.
15:50–16:50	Time series	Heart-rate zones as % of HR_{max} (220 − age), time per zone, `rolling`, `resample`.
16:50–17:30	Mini-case in pairs	A training report: one row per athlete.

Exercises: (0) recap: repair the broken cell; (1) group by gender and age group; (2) what did removing the duplicates cost?; (3) progress in jump height; (4) minutes in zone 4–5 per athlete;

mini-case: training report with the minutes in zone 4–5, the maximum speed and the distance covered per athlete.

7 Working method

- Each block starts with a short demo in the notebook, followed by exercises you solve in the notebook itself.
- Use the automatic checks as feedback, not as the goal: when a check turns green, make sure you can explain *why* your code works.
- You do the mini-case at the end in pairs.

8 Preparation

- Check before the session that you can open the notebook, locally or on marimo.app.
- Make sure you can explain every step of the cleaning pipeline at the end of lab 3, because lab 4 starts from it.
- Read the course text for lab 4 next to the notebook.